SHANMUGASUNDARAM M

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Education

Chennai Institute of Technology 2023 – 2027
B.Tech in Computer Science and Business Systems **CGPA: 8.50**

Experience

Project Trainee (Zoho) Sep 2026 – Oct 2026

- Studied Jakarta EE fundamentals and enterprise Java development concepts, gaining exposure to server-side application architecture and enterprise software development practices.
- Practiced Git version control workflows, including branching, merging, and commit management, while exploring Zoho's enterprise product ecosystem and business application workflows.

Software Development Intern (Mobile Application Development COE – CIT) Nov 2025 – Jan 2026

- Developed and delivered 15+ RESTful API endpoints using FastAPI for a campus management platform, implementing CRUD operations, backend business logic, and student and administrator workflows.
- Conducted API testing and request-response validation using Postman, collaborating with developers to troubleshoot functional issues, verify endpoint behavior, and improve backend integration and reliability.

Projects

SurveyGuard
Python, FastAPI, React, TypeScript, Machine Learning

- Developed a privacy-preserving survey management platform with AI-powered survey generation, role-based workflows, data collection, and analytics dashboards using FastAPI and React.
- Implemented IsolationForest-based fraud detection and a synthetic-fraud evaluation harness to assess detection performance using precision, recall, and F1-score.
- Built privacy-preserving data processing workflows incorporating k-anonymity, l-diversity, PII redaction with Microsoft Presidio, and differential privacy experiments using IBM OpenDP.

AIS (Academic Intelligence System)
React, TypeScript, FastAPI, Redis, Celery, ChromaDB, Garage, OAuth 2.0

- Developed a full-stack academic platform using React and FastAPI, with Redis-backed caching and Celery task queues for asynchronous document processing.
- Implemented embedding-based semantic search with ChromaDB to retrieve relevant academic resources from unstructured educational data.

Campus Connect
Python, FastAPI, SQLAlchemy, MySQL, JWT

- Designed relational database schemas and developed RESTful APIs using FastAPI and SQLAlchemy for student management, attendance tracking, authentication, and administrative workflows.
- Implemented JWT-based role-based access control (RBAC) across 30+ endpoints to enforce role-specific permissions and protect student and administrative resources.

Skills

Languages: Python, Java, C++, JavaScript, TypeScript
Backend: FastAPI,Node Js, Express.js, Jakarta EE, REST APIs, gRPC, Celery
Databases & Caching: MySQL, MongoDB, Redis, ChromaDB, Garage
AI & ML: scikit-learn, PyTorch, LangChain, LangGraph, MCP
Frontend: React.js, Tailwind CSS, TanStack Query
Cloud & Tools: AWS, Docker, Linux, Git, Bash, GitHub, cURL
Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Machine Learning

Achievements & Problem Solving

- **GATE CSE 2026:** Qualified with a score of 387.
- **LeetCode:** Solved 721+ problems; maximum contest rating of 1702.
- **HackChrono Hackathon, Chandigarh University:** Finalist.
- **CITIL SDG Hackathon, Chennai Institute of Technology:** Finalist.

Certifications

- Cisco CCNA 1, 2, 3: Networking, Switching/Routing, Enterprise Security & Automation
- Coursera: Google Cloud Fundamentals; Networking in Google Cloud